6 June 1955

SUPERSEDING
NAVORD OS 1895C (in part)*
18 FEBRUARY 1955

MILITARY SPECIFICATION

PRIMER, ELECTRIC, MARK 44 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading, and preparation for delivery of Electric Primer Mark 44 Mod 0 and the methods of inspection and tests upon which acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, drawings and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

LLL-B-631	-- Boxes; Fiber Corrugated (For Domestic Shipment).

MILITARY

MIL-P-116	-- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	-- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-- Marking for Shipment and Storage.
MIL-STD-353	-- Transportation Vibration Test for Use in Production of Fuzes.

DRAWINGS

BUREAU OF ORDNANCE

LD No. 168789	-- Primer Container Mk 7 Mod 0 (Outer).
LD No. 276294	-- Primer (Electric) Mk 44 Mod 0.
397798	-- Machine, Jolt Testing for Assembled Fuzes (Assembly).

And all documents listed on the above LD's and drawings.

* (See 6.2.3)

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PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives, and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950)".)

3. REQUIREMENTS

3.1 General.

3.1.1 The primers shall be manufactured in accordance with BuOrd LD 276294 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Bridge wire subassembly, continuity. -- Before assembling the bridge wire subassembly into the ignition element subassembly, it shall be thoroughly cleaned to remove solder flux and contamination. After cleaning, the bridge wire shall be flashed by application of an electric current to assure continuity of the bridge wire. The current shall be regulated to produce a

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visible red glow of the bridges. An EMF of approximately 1.5 volts a-c or d-c may be used with a suitable series resistor to adapt the generator characteristics to the primer resistance. Flashing is a necessary manufacturing operation and shall be conducted on each assembly. Assemblies which fail to glow in both legs of the bridge shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridge.

3.3 Primer resistance. -- When tested electrically, as specified in 4.6.1, the resistance shall be in accordance with the applicable drawing.

3.4 Electrical functioning. -- The primers shall fire within 0.006 second after the stock and electrode are subjected to a d-c potential difference of 20 volts. Functioning shall not result in blowback, split stock or the detachment of metallic components of the primer. The provisions of 4.6.2 shall be applicable for establishing conformance.

3.5 Jolt test. -- Each lot of primers shall pass the Jolt Test specified in 4.6.3.

3.6 Vibration. -- The primers shall be safe and operable after vibration in accordance with the provisions of Standard MIL-STD-353 and 4.6.4.

3.7 Drop test. -- Each lot of primers shall pass the drop test specified in 4.6.5.

3.8 Moisture. -- The moisture content of the black powder shall not exceed 0.70 percent.

3.9 Subassemblies. -- Component parts of subassemblies shown on applicable drawings shall be thoroughly cleaned and dried before assembly. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent which is employed. All assembly operations shall be conducted in a careful manner to prevent inclusion of moisture, oils and foreign matter. Where parts involved are of plastic or fiber, such parts after final assembly and machining shall be protected from moisture or other influence which would affect dimensional stability.

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3.10 Assembly.

3.10.1 Priming composition. -- The priming composition shall be compounded by an approved method which maintains the weight composition of non-volatile constituents prescribed by the drawing. The finished composition when ready for buttering into the bridge wire assembly shall be a uniformly mixed slurry in which all of the particles are wetted. The consistency shall be such that the liquid will not separate from the mixture when the dish is tilted. Adjustment of the consistency may be obtained by the addition of pure butyl acetate or by evaporation of the solvent. Any adjustment shall be followed by thorough remixing.

3.10.2 Loading of ignition charge. -- The technique of wet loading of the ignition charge shall be such as to avoid damage to the bridge wire. The priming composition, when consolidated to assure freedom from voids, shall be smoothed off flush with the top of the cavity. All priming composition shall be removed from the exterior lateral areas of the inner cup by means of a clean cloth wetted with butyl acetate. The charged igniter shall be dried at a temperature of 120 degrees F. for a minimum period of 16 hours. At least four hours of drying shall immediately precede assembly of the igniter into the ignition element subassembly. The dried ignition charge shall be free from cracks, voids, and separation from the inner cup. The charge shall entirely cover the bridge wire. The exterior cylindrical surface of the inner cup shall be free from priming composition.

3.11 Weight. -- The loading plant shall check-weigh each loaded primer to insure presence of the black powder charge.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1 Lot. -- As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purposes.

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4.2 Preproduction sample. -- Before entering into quantity production a preproduction sample of 300 primers shall be delivered by the contractor for test to an agency designated by the Government. These primers shall be manufactured by the same procedures and processes and at the same location proposed by the contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the contractor will be authorized by the Government to proceed with the submission of the product for acceptance. Accepted samples shall become the property of the government and shall be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for subsequent contract. -- The necessity for a preproduction sample will be determined by the Government, when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Inspection. -- The general inspection procedures shall be in accordance with General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OP 400; Government Inspection shall be in accordance with the provisions of General Procedure for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.4 Production lot. -- For procurement purposes the production lot shall consist of 20,000 loaded primers plus those required for all test purposes. Unless otherwise stated in the contract, order, or requisition, all units for test purposes shall be provided at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4.1.1.

4.5 Test equipment.

4.5.1 Jolt, Testing Machine shown on Bureau of Ordnance Drawing 397798.

4.6 Tests.

4.6.1 Electrical characteristics. -- Conformance with the bridge circuit resistance of the drawing shall be ascertained by means of a test circuit which limits the bridge current to 50 milliamperes maximum. A safety chamber with an interlock switch shall be used to protect the operator during

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application of the current. Primers which conform to the resistance requirements of the applicable drawing shall be acceptable. The insulator shall be firmly held by the stock crimp and shall not be broken or displaced.

4.6.2 Electrical functioning. -- The primers shall be tested in a safety chamber which simulates support against the forces of explosion equivalent to service operation. Apply a d-c potential of 20.0 volts using the primer stock and electrode as terminals. It is important that firm contact on clean bright surfaces be obtained to avoid high contact resistance. The d-c voltage shall be supplied by an essentially noninductive source of 2.0 ohms internal resistance or by means of a circuit of equivalent characteristics. The test shall be conducted in accordance with Standard MIL-STD-105, Multiple Sampling Plan, Inspection Level I, A.Q.L. 0.40 percent defective. Electronic timing which is actuated by the firing impulse and the explosive output shall be used. The explosive output shall be ascertained at the venthole nearest to the stock by means of an ionization probe or a device with at least equal response speed. Primers which conform to the requirements of 3.4 shall be acceptable under this test.

4.6.3 Jolt test. -- Twenty-five primers from each of the first five production lots of loaded primers shall be subjected to the standard jolt test in which each primer shall be securely supported in such a manner as to be protected from injury or from direct impact but will receive the full impact of the jolt arm. The jolt test machine used for this test should be similar to that shown on Bureau of Ordnance Drawing 397798. The primers shall be jolted 600 times vertically with the extension tube up, and 600 times vertically with the extension tube down. No primer shall fire and there shall be no mechanical breakdown which would render the primer unsafe to handle or dangerous to use as determined by examination.

4.6.4 Vibration test. -- Twenty-four (24) primers from each production lot shall be rigidly mounted to the vibration table which is specified in Standard MIL-STD-353. The sample orientations for the three (3) phase test shall be axis-vertical closing disc-up, axis-vertical closing disc-down, and axis-horizontal. Subject the primers to the vibration schedule of Standard MIL-STD-353. The lot shall be rejected if one (1) of the primers explodes or is unsafe to handle and use for service purposes.

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The lot shall be rejected if more than one (1) primer fails to meet the bridge resistance requirements of the drawing. Subject the samples to the electrical functioning test of 4.6.2. The incidence of more than one (1) functioning defect shall cause rejection of the lot. The incidence of one (1) defective shall cause a retest on the basis of a sample of 48 units with rejection of the lot on obtaining more than one (1) defective.

4.6.5 Drop test. -- Twenty-five primers from each of the first five production lots and ten from each succeeding lot shall be given a 40-foot drop test onto steel plate set in a solid foundation. Each primer shall be secured in a fixture so as to be supported by the stock threads and dropped with the extension tube down. The supporting fixture and not the primer shall strike the steel plate. To pass this test none of the primers shall fire and the primer shall be safe to handle and dispose of.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Inert primers or components. -- Unless otherwise specified in the contract or order, inert primers or components shall be packaged in commercial fiberboard containers meeting the requirements of Federal Specification LLL-B-631. Each container shall incorporate corrugated or solid fiberboard dividers or collars suitable for preventing damage to the threaded areas in handling and transportation. The gross weight of the container and contents shall not exceed 70 pounds.

5.1.2 Loaded primers. -- Unless otherwise specified, ten (10) Primers Mk 44 Mod 0 shall be unit packaged Method IIe (less desiccant) conforming to the requirements of Military Specification MIL-P-116. Corrugated fiberboard trays (with suitable mounting holes) shall be employed to support and prevent movement of contents.

5.2 Packing. -- Multiples of unit packages, as described above in paragraph 5.1.2 shall be packed in "Primer Container Mk 7 Mod 0 (Outer)", Bureau of Ordnance List of Drawings LD No. 168789.

5.3 Marking. -- In addition to any special markings required by the contract or order, all markings shall be in accordance with Standard MIL-STD-129, "Marking of Shipments" and Code of Federal Regulations 49-CFR-71-78.

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5.4 Data card. -- Data cards made from commercial white cardboard and printed as shown in Figure 1 shall be properly filled out for each primer lot. One card shall be placed in the top of each shipping box, two shall accompany ballistic samples, five shall be forwarded to the Government, and twenty shall go to the Ammunition Depot receiving the primers.

FIGURE 1

Primer Data Card

Type _____ Lot No. _____

Project Order No. _____ Lot Size _____

Loaded by _____ Date _____

Inspected by _____ Commanding Officer _____

Components	Manufacturer	Lot No.	Year Mfr.
Inert parts set			
Black Powder FFG			
Black Powder Cannon			
Priming Composition			

6. NOTES

6.1 Intended use. -- Primer Mk 44 Mod 0 is intended as a test primer for use when testing the firing circuits of all calibers of case guns from 3 inch through 8 inch inclusive.

6.2 Advisory procedures for the preparation, loading and handling of the explosive components.

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6.2.1 Preparation of priming composition. -- A small amount, approximately ten milliliters of lead styphnate, as received, is placed in a tared conductive rubber dish. The styphnate is allowed to settle, and the supernatant water is removed by decantation or by siphoning. Approximately 25 milliliters of methanol are added and the mixture stirred gently, allowed to settle, and siphoned. Approximately 25 milliliters of ether are added and the stirring, settling and siphoning procedure repeated. The remaining mixture is air-dried at ambient temperature for two hours and then oven-dried for 12 hours at 50 degrees C. The dry weight of the styphnate is then determined. In a separate container, dry cellulose nitrate, equivalent to 2.0 percent of the dry weight of the lead styphnate, is mixed with butyl acetate to make an approximately ½ percent solution. (A weighed quantity of gravimetrically standardized stock solution of approximately ½ percent cellulose nitrate concentration may be used to obtain the 2.0 percent of cellulose nitrate required.) This solution is added to the dry styphnate and mixed until the styphnate is thoroughly wetted.

6.2.2 Loading and handling lead styphnate priming composition. -- The loading room should have a relative humidity of 70 to 80 percent. The moisture content should be maintained at all times. Only approximately two grams of priming composition should be present at each loading station and only when an operator is present. Any composition which is spilled should be cleaned up immediately by means of a cloth wet with butyl acetate; the soiled cloth should be placed in a special disposal can. The mixing and loading of the priming composition should be done behind an approved safety shield whenever possible. All fixtures, equipment, benches, floors, etc., should be properly grounded to prevent accidental ignition due to static discharges, electrical arcing, or sparking. Butyl acetate confers a degree of resistance to explosion and ignition by impact and friction. However, it is emphasized that the resistance to explosion obtained by wetting with butyl acetate is of a lower order than that obtained by water wetting. Butyl acetate is inflammable and can produce vapors which may be ignited at dry-house temperatures. The loading dish should be constantly checked to prevent the mixture from becoming too dry, particularly around the edges and on the top of the dish. If necessary, the operator should add butyl acetate and work the mixture to the proper buttering consistency. To charge the bridge wire subassembly with the ignition charge, convey sufficient composition on a non-sparking spatula to butter (load) one unit. Scrape off, and press the composition into the inner

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cup cavity using short strokes, downward and outward. Excess composition should be scraped off flush with the surface of the inner cup. The bridge wire should not be visible. Clean off all excess mixture from the sides of the inner cup using clean muslin cloth, wet with butyl acetate. Lead styphnate is characterized by unusual sensitivity to electrical discharges. Precautions such as grounding and cleanliness of tools and utensils should be observed.

6.2.3 Superseding documents. -- This specification supersedes that part of NAVORD OS 1895 applicable to Electric Primer Mark 44 Mod 0. Specifications which supersede parts of NAVORD OS 1895 applicable to other primers are as follows:

MIL-P-18714 (NOrd) -- Primer, Electric, Mk 45 Mod 0
MIL-P-18715 (NOrd) -- Primer, Electric, Mk 46 Mod 0
MIL-P-18716 (NOrd) -- Primer, Electric, Mk 48 Mods 0 & 1

Patent Notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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